

## **GUIDED MEASUREMENT SOFTWARE AND USER INTERACTION EMBODIMENTS**

In various embodiments, operation of the intraluminal compliance and distance profiling device is supported by guided measurement software configured to assist users in performing repeatable and validated measurement sessions. Guided interaction may be implemented through mobile applications, dedicated computing interfaces, embedded displays, or remote software environments.

The software system may provide step-by-step procedural guidance including insertion instructions, stabilization prompts, confirmation of proper positioning, and initiation of sensing sequences. Guidance may be delivered through visual interfaces, auditory cues, haptic feedback, or combinations thereof.

In certain embodiments, the software monitors real-time sensor data to determine whether measurement conditions meet predefined acceptance criteria. If instability, excessive motion, or improper positioning is detected, the system may instruct the user to adjust positioning before continuing measurement.

Guided workflows may include timed stabilization periods during which sensor readings are evaluated for consistency. Measurement acquisition may automatically begin once stability thresholds are satisfied, thereby reducing user-dependent variability.

The software may dynamically adjust workflow steps based on device configuration, selected anatomy mode, or historical user performance. Adaptive guidance may shorten procedures for experienced users or provide expanded assistance for new users.

In embodiments including inflatable compliance measurement, the software may control or supervise inflation sequences, display pressure progression, and provide safety notifications when approaching predefined limits. Users may be guided through incremental expansion stages to ensure safe and consistent operation.

User interaction interfaces may include progress indicators, completion scores, session summaries, or confidence ratings representing measurement quality. Such feedback may encourage proper usage behavior and improve longitudinal data reliability.

The guided measurement system may further incorporate onboarding tutorials, instructional simulations, or training modes designed to familiarize users with device handling and safety procedures prior to first measurement.

In certain embodiments, software workflows integrate directly with computational profiling systems, automatically transferring validated measurement results into a persistent user profile. Session metadata such as timestamp, configuration mode, calibration state, and confidence score may be stored alongside measured values.

Accordingly, guided measurement software embodiments reduce operational variability, improve measurement accuracy, enhance safety, and enable consistent data acquisition across diverse users and deployment environments.